

"KartGo – Queue-Free Smart Grocery Shopping System Using Mobile Application and RFID Technology"

A Seminar Report

Submitted in partial fulfilment of the requirements for the Degree of

Bachelor of Technology

in

INFORMATION TECHNOLOGY

by

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CERTIFICATE

This is to certify that the Seminar entitled “KartGo – Queue-Free Smart Grocery Shopping System Using Mobile Application and RFID Technology” is a genuine record of work undertaken by

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under my supervision and guidance, submitted in partial fulfilment of the degree of Bachelor of Technology in Information and Technology at Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded (M.S.), INDIA.

The findings and content contained in this report have not been presented to any other university or institution for the purpose of obtaining a degree or diploma.

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DECLARATION

I hereby declare that the Seminar Report titled "KartGo – Queue-Free Smart Grocery Shopping System Using Mobile Application and RFID Technology", submitted in partial fulfilment of the requirements for the degree of Bachelor of Technology in Information Technology at SGGS IE&T, Nanded, represents my own original and independent work, carried out under the supervision of Mr Prem Agrawal . The content of this report has not been submitted to any other university or institution for any degree or diploma. All references and sources have been appropriately acknowledged.

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ABSTRACT

Title of Project: KartGo – Queue-Free Smart Grocery Shopping System Using Mobile Application and RFID Technology"

Datasets and Models:

The system uses a structured product catalogue of over 15 items incorporating real Indian UPC/EAN barcodes, product names, categories, prices, and brand information. Design models include: a Use Case Diagram covering three actors, a Level-0 Context DFD, a Level-1 Detailed DFD with five processes, an Entity-Relationship Diagram, and a three-tier System Architecture Diagram."

System Requirements:

Hardware requirements include an Android or iOS smartphone with a rear-facing camera of at least 5 MP, passive RFID tags operating at 13.56 MHz, and an RFID exit gate reader. Software requirements include Chrome 80 or above or Safari 14 or above, HTTPS hosting via Netlify, a Supabase backend comprising PostgreSQL along with Supabase Auth and Twilio SMS, QuaggaJS v0.12.1, QRCode.js v1.0.0, and the Supabase JS SDK v2."

Description:

KartGo is a mobile-first Progressive Web Application built to remove checkout queues from Indian supermarkets by allowing shoppers to scan product barcodes using their own smartphone camera while they move through the store. At the time of payment, the application produces a 5-minute OTP along with a payment QR code. A store guard at the exit performs a spot-check on item count, grants approval to the customer, and RFID tags on all purchased items are then deactivated. A conventional cashier counter is maintained for customers who do not own smartphones."

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
DFD	Data Flow Diagram
EAN	European Article Number
HTTPS	HyperText Transfer Protocol Secure
IoT	Internet of Things
JSON	JavaScript Object Notation
OTP	One-Time Password
PWA	Progressive Web Application
QR	Quick Response
RFID	Radio Frequency Identification
RLS	Row-Level Security
SMS	Short Message Service
SQL	Structured Query Language
UPC	Universal Product Code
UPI	Unified Payments Interface
UUID	Universally Unique Identifier
UX	User Experience

CHAPTER 1: INTRODUCTION"

1.1 Background and Motivation"

India's retail industry stands among the most rapidly expanding in the world, with projections indicating a market size of USD 1.7 trillion by 2026 (India Brand Equity Foundation, 2024). The organised segment of this industry — spanning supermarkets, hypermarkets, and department stores — has recorded compounded annual growth of over 25% during the past ten years. Well-established retail chains such as DMart, Reliance Fresh, Big Bazaar, Spencer's, and Star Bazaar together cater to tens of millions of shoppers each week across urban and semi-urban India. This growth has been driven by a combination of increasing urbanisation, rising household incomes, a growing aspirational middle class, and enhanced supply chain infrastructure that has made fresh produce more widely accessible."

Yet one critical operational problem has remained largely unresolved despite this expansion: congestion at the checkout counter. During busy periods — evenings on working days, weekends, and major festive occasions such as Diwali, Navratri, and Holi — queues at popular supermarkets routinely make customers wait anywhere between 20 and 45 minutes. A study conducted by Sharma and Mehta (2018) across 12 DMart stores in Maharashtra recorded an average peak-hour wait of 23.4 minutes for customers carrying 15 or more items. The same study found that 34% of cart abandonments happened primarily because of queue length, with the resulting annual revenue loss estimated at INR 4,200 crore across the organised sector. Repeated consumer surveys in India consistently identify checkout waiting time as the single biggest negative factor in the supermarket experience."

Several countries have already tackled this challenge. Japan and Indonesia, for instance, have deployed smart shopping cart systems that let customers scan items as they shop, cutting checkout time by 40 to 50 percent. Major Japanese retail chains, including AEON and Ito-Yokado, issue dedicated handheld RFID scanners to shoppers at the entrance. In India, however, distributing store-owned hardware is not feasible: a single popular DMart outlet may attract 3,000 to 5,000 customers in a day, making procurement costs and hygiene concerns around shared devices prohibitive. Given that India had over 750 million smartphone users as of 2025 — a figure growing at around 6% annually — a far more practical approach is to use the customer's own mobile device as the scanning tool. This core insight forms the foundation of KartGo."

The system draws direct inspiration from an RFID smart cart study carried out at Ranch Market and Food Hall in Jakarta by Wibowo et al. (2021), which recorded a 48.2% drop in checkout processing time using an Arduino-based RFID scanning setup. KartGo adapts and expands this concept for Indian conditions by substituting dedicated hardware with a smartphone Progressive Web Application, while introducing a procedural anti-theft mechanism. This addition was specifically designed to address the shrinkage problems that led Walmart to shut down its comparable Scan and Go pilot in 2019."

1.2 Problem Statement"

The conventional supermarket checkout creates a single concentrated bottleneck at the end of every shopping trip. No matter how quickly a customer moves through the aisles, every purchased item must pass through a cashier scan when shopping ends. This structure introduces three compounding inefficiencies. First, all scanning gets pushed into a brief window at the end rather than being spread across the time spent shopping. Second, the cashier becomes a single point of failure whose absence or reduced speed immediately slows service for everyone

waiting behind. Third, self-checkout kiosks — the conventional alternative — carry a capital cost of INR 3 to 5 lakh per unit, encounter weight-sensor errors in 8 to 12 percent of transactions that require staff involvement, and are poorly suited to elderly shoppers who make up a significant share of peak-hour customers."

The core problem can therefore be stated as follows: How can the checkout waiting time in Indian supermarkets be reduced by at least 30 percent, without requiring any per-customer hardware investment by the store, while maintaining adequate theft deterrence and providing a usable path for customers who do not own smartphones?"

1.3 Objectives of the Project"

Objective 1: To develop a cross-platform Progressive Web Application that allows customers to scan product barcodes using their personal smartphone camera, with no app store download, installation, or device-specific software needed."

Objective 2: To build a dependable barcode scanning module capable of handling UPC-A, EAN-13, EAN-8, and Code-128 formats through QuaggaJS, along with three alternative input modes — product tap-list, QR code scanning, and manual entry — to handle situations where camera scanning is not practical."

Objective 3: To create a secure 5-minute OTP checkout mechanism featuring an animated SVG countdown ring, a payment QR code that encodes complete session data, and Supabase session synchronisation to generate a verifiable proof-of-shopping token."

Objective 4: To develop a Guard Verification Module enabling store staff to verify item counts, perform randomised spot-checks on selected items, and either approve or flag transactions — forming a procedural anti-theft layer complementary to the store's existing RFID infrastructure."

Objective 5: To integrate a Supabase backend supporting phone-number OTP authentication, real-time cart session synchronisation, persistent shopping history, and budget tracking — all protected by row-level security policies that enforce strict user data isolation."

Objective 6: To provide a Traditional Queue Module with token number assignment and live status display for customers without smartphones, and to generate a printable QR code grid for shelf labelling."

1.4 Scope of the Project"

This project covers the complete design, development, and testing of the KartGo PWA and its Supabase backend. The application handles the entire customer journey: phone OTP login, product scanning through four different modes, cart management with budget tracking, a 5-minute OTP checkout with QR code, guard verification including spot-check, RFID clearance signalling, traditional queue management for non-smartphone users, shopping history, and user profile management. The system includes a catalogue of 15 representative Indian grocery products with real UPC and EAN barcodes and is intended for single-store deployment."

Physical RFID gate integration is represented at the software level through a database flag; the store-side service that monitors this flag and sends physical deactivation signals to the hardware falls outside the scope of this academic implementation. Similarly, actual payment processing is out of scope — the OTP acts as a checkout verification token at a lightly staffed payment station, while financial transactions are handled by the store's existing UPI and card infrastructure."

Table 1.1 compares KartGo with existing checkout approaches across five key dimensions:"

Dimension	Traditional Cashier	Self-Checkout Kiosk	Amazon Go	Wibowo RFID Smart Cart	KartGo (This Project)
Per-Customer Hardware	None (cashier uses store hardware)	INR 3–5 lakh kiosk	USD 1–3M per store	Arduino + RFID cart	Customer's own smartphone
Checkout Time Reduction	0% (baseline)	20–30%	~100% (walk out)	48.2%	71.6–79.8%
Anti-Theft Mechanism	Cashier verification	Weight sensor	Computer vision	RFID + cashier	Guard count + RFID + spot-check
Indian Deployment Feasibility	High (existing)	Low (cost, errors)	Very Low (infra cost)	Medium (hardware cost)	High (zero extra hardware)
Smartphone Dependency	None	None	None	None	Yes (traditional queue fallback provided)
Internet Required	No	No	No	Yes (Wi-Fi for cart)	Yes (offline cart fallback provided)

Table 1.1: Comparison of Checkout Approaches Across Key Dimensions"

1.5 Organisation of the Report"

The rest of this report is arranged as follows. Chapter 2 provides a thorough literature review covering queue management theory, self-checkout and scan-as-you-shop systems, RFID applications in retail, Progressive Web Applications, barcode scanning libraries, and the digital transformation of Indian retail. Chapter 3 covers system analysis and design, including feasibility analysis, functional and non-functional requirements, three-tier architecture, use case diagram, data flow diagram, entity-relationship model, and database schema. Chapter 4 describes the implementation of every system module in technical detail, covering the technology stack, PWA architecture, authentication flow, barcode scanning, cart management, OTP checkout, guard verification, queue management, budget tracking, and a consolidated module summary table. Chapter 5 presents the testing approach, unit and integration test outcomes, performance benchmarks across four devices under three lighting conditions and five barcode types, checkout time comparisons, and identified limitations. Chapter 6 concludes with a summary of contributions and eight specific directions for future development. A full reference list appears at the end."

Figure 1.1: KartGo Customer Journey Flowchart"

CHAPTER 2: LITERATURE REVIEW"

2.1 Queue Management Systems"

Queue management in retail settings has been a well-studied area in both operations research and service management. The M/M/c queueing model derived from Erlang's foundational work on telephone exchanges remains the most commonly applied framework for analysing supermarket checkout behaviour. In this model, customer arrivals follow a Poisson process at rate λ , service times are exponentially distributed with mean $1/\mu$, and c cashiers operate as parallel servers. During peak hours, Indian supermarkets regularly enter the heavy-traffic regime — where the server utilisation ratio $\rho = \lambda/c\mu$ approaches 1 — causing average waiting times to grow steeply."

Bitner (1992) studied how the physical layout of service environments influences customer perceptions and demonstrated that the visibility of a queue and the sense of fairness it creates significantly affect satisfaction, independently of how long someone actually waits. Shoppers who can see where a queue ends express considerably higher satisfaction than those stuck in queues that extend around corners, even when actual wait durations are identical. This finding directly shaped KartGo's Traditional Queue module, which shows each customer their live position in the queue and an estimated wait time. Sharma and Mehta (2018) placed a monetary figure on this problem in the Indian context: their survey across 12 DMart outlets found that cart abandonment linked to checkout queues costs organised retail approximately INR 4,200 crore each year. While queue management tools deployed by Walmart India and Reliance Retail — which use sensor-based alerts to open additional cashier counters — can trim average wait times by 15 to 20 percent, they leave the underlying checkout architecture untouched."

2.2 Self-Checkout and Scan-as-You-Shop Systems"

Self-checkout kiosks were commercially introduced by Kroger in the United States in 1992 and are now in use at more than 400,000 locations worldwide (NCR Corporation, 2023). While these units remove the cashier from the scanning task, they still require all items to be scanned after shopping is complete, preserving the end-of-trip bottleneck. Furthermore, weight-sensor failures occur in 8 to 12 percent of transactions, calling for staff intervention and eroding much of the time saved. In the Indian context, the capital cost of INR 3 to 5 lakh per unit, combined with floor space constraints and limited uptake among elderly shoppers, makes large-scale kiosk deployment economically difficult for most retailers working on net margins of just 3 to 6 percent."

The scan-as-you-shop concept takes a more fundamental approach to the problem by distributing scanning across the full shopping trip rather than concentrating it at the end. Amazon Go, launched in Seattle in January 2018, pushes this idea to its logical conclusion through a combination of ceiling-mounted cameras, shelf weight sensors, and deep learning algorithms that remove checkout entirely. Its infrastructure costs — estimated at USD 1 to 3 million per store — have restricted deployment to small-format convenience outlets. Walmart's Scan and Go mobile pilot (2018–2019) is the closest prior precedent to KartGo: shoppers used personal smartphones to scan items and then paid at a dedicated express lane, achieving checkout time reductions of around 40 percent. The programme was nonetheless discontinued after theft rates were found to be 16 times higher than at standard cashier lanes — a finding that directly influenced KartGo's design of the guard verification and random spot-check features."

The most directly relevant reference for this project is Wibowo et al. (2021), who deployed an RFID-integrated smart cart system at Ranch Market and Food Hall in Grand Indonesia Mall, Jakarta, between November 2019 and January 2020. Their system combined an Arduino Uno microcontroller, an RC522 RFID reader operating at 13.56 MHz with a 5 cm read range, and an ESP8266 Wi-Fi module, paired with a Flutter-based customer app and a Node.js/PostgreSQL/React cashier interface. Experiments conducted with 10 customers handling a combined total of 100 items showed that checkout time fell from 6,255 seconds using the traditional method to 3,240 seconds with the smart cart — a 48.2 percent improvement. These results form the primary quantitative benchmark for evaluating KartGo's performance. Tesco's Scan and Shop programme in the UK, which issues handheld scanners to customers and re-scans 5 to 10 percent of baskets at random, served as the reference model for designing KartGo's guard verification step."

Figure 2.1: Checkout Time Comparison: Traditional vs RFID Smart Cart (Wibowo et al., 2021)"

2.3 RFID Technology in Retail Applications"

Radio Frequency Identification (RFID) uses radio-wave-based electromagnetic fields to identify and track tags attached to physical objects. A complete RFID system has three components: a passive tag consisting of a microchip and an antenna that draws power from the reader's field rather than a battery; a reader that emits radio waves and interprets tag responses; and a backend system that links tag identifiers to product records in a database. Retail deployments primarily use two frequency bands: High Frequency (13.56 MHz, ISO 15693, with read ranges of 10 to 50 cm) for point-of-sale and anti-theft applications, and Ultra High Frequency (860 to 960 MHz, EPC Gen2, with read ranges of 1 to 12 m) for supply chain and inventory management. Electronic Article Surveillance (EAS) anti-theft systems typically operate at acousto-magnetic (8.2 MHz) or electromagnetic (58 kHz) frequencies."

Research by Rekik, Syntetos, and Gaalman (2008) showed that deploying RFID reduced inventory record errors in retail stores by as much as 80 percent, directly improving shelf availability and lowering the costs associated with overstock. Ustundag and Tanyas (2009) modelled the supply chain savings from RFID and determined that the break-even point occurs when tag prices fall below USD 0.08 — a threshold now surpassed, with UHF tags available in bulk for under USD 0.05. Bhatt and Bhatt (2012) reviewed RFID usage in Indian retail and found that adoption was concentrated in warehouse and supply chain functions, with integration at the point of sale still lagging. KartGo addresses this gap by treating the anti-theft RFID tags already attached to most packaged goods at the manufacturer level as the deactivation targets, removing the need for stores to purchase or apply new tags. Within the KartGo architecture, the Supabase database mediates between the software and the hardware: the `rfid_cleared` flag acts as an API contract that allows any RFID hardware vendor's system to integrate without requiring changes to the application code."

2.4 Progressive Web Applications and Barcode Scanning"

Progressive Web Applications (PWAs) bring native app capabilities to the browser by using service workers for offline support and background synchronisation, web app manifests for home screen installation and standalone display, and modern device APIs for camera access, push notifications, and haptic feedback. The concept was introduced by Google's Alex Russell in 2015, and the core standards were formalised by the W3C and adopted across all major browsers by 2019. For Indian retail, the zero-download approach is particularly important: a shopper can launch the full application simply by scanning a QR code at the store entrance,

needing no app store account, no free storage space, and no installation time. The commercial case is equally compelling: Flipkart Lite's PWA delivered 70 percent higher conversions along with a 3× reduction in data consumption; AliExpress saw a 104 percent improvement in conversions among first-time users; Forbes recorded a 6× increase in conversion rates."

In-browser barcode scanning relies on the `MediaDevices.getUserMedia()` API for camera access and computer vision libraries for detection. KartGo uses QuaggaJS, which processes video frames through a sequence of steps: Gaussian blur to reduce noise, Sobel edge detection to compute intensity gradients, a Hough transform to identify lines, and Freeman chain code analysis to recognise barcode patterns. Because all processing happens on the client side, the scanner functions without any network connection. A benchmarking study by Gallo et al. (2022) comparing five barcode libraries found that QuaggaJS achieved a detection success rate of 91.3 percent at an average latency of 0.73 seconds on mid-range devices; ZXing-js performed slightly better at 93.1 percent but was slower at 1.12 seconds; and native Android ML Kit led at 98.2 percent with only 0.31-second latency but is not accessible from within a browser. QuaggaJS therefore represents the best balance of detection performance, zero licensing cost, and cross-platform support — including iOS Safari, where ML Kit is unavailable."

2.5 Summary and Research Gap"

The body of published literature covers queue management theory, self-checkout technologies, RFID in retail, and PWA development comprehensively. However, no prior system has been purpose-built for the Indian supermarket context — one that combines high daily customer volumes that rule out per-customer hardware, a wide diversity of smartphone models that demands robust cross-platform compatibility, thin retail margins that require near-zero infrastructure cost, and pre-existing RFID anti-theft tags on merchandise that represent an integration opportunity without fresh hardware investment. KartGo fills this gap by bringing together PWA technology, QuaggaJS barcode scanning, a Supabase cloud backend, and a procedural guard-verification anti-theft mechanism into one cohesive system calibrated for Indian deployment conditions. In doing so, it simultaneously addresses the two most critical weaknesses identified in the existing literature: the shrinkage problem that ended Walmart's Scan and Go pilot, and the hardware cost barrier that limits the scalability of the Wibowo et al. RFID cart approach."

2.6 Indian Retail Digital Transformation Context"

Appreciating the broader digital transformation of Indian retail is necessary to understand why KartGo is particularly well-timed. India's Unified Payments Interface (UPI), launched by the National Payments Corporation of India (NPCI) in 2016, processed more than 100 billion transactions worth INR 180 lakh crore in 2024, establishing India as the world's largest real-time payments market by volume. The widespread adoption of UPI means that Indian consumers are already thoroughly familiar with scanning QR codes to pay — a behaviour that directly lowers the learning curve for KartGo's QR-based checkout verification."

The government's Digital India programme has accelerated broadband rollout, with 4G and 5G coverage now reaching over 95 percent of urban areas and expanding into semi-urban regions. Reliance Jio's entry into the mobile data market in 2016 drove prices to among the lowest in the world — approximately INR 10 per gigabyte — making data-intensive applications like KartGo economically accessible even to lower-income shoppers. The ONDC (Open Network for Digital Commerce) initiative, launched in 2022, is building standardised retail APIs that

future versions of KartGo could integrate with for product discovery and inventory synchronisation."

Singh and Gupta (2023) examined smartphone ownership and digital payment habits across urban Indian demographic groups and found that even among shoppers aged 60 and above in cities with populations exceeding one million, smartphone ownership was above 42 percent and near-universal familiarity with QR-based payment apps was recorded among smartphone users. This evidence supports the viability of KartGo's smartphone checkout for the majority of urban Indian supermarket visitors, while also justifying the traditional queue fallback that remains available for the 58 percent of older shoppers in that demographic who do not yet own a smartphone."

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN"

3.1 Feasibility Analysis"

Technical Feasibility: Every component used in KartGo is mature, widely deployed, and battle-tested in production environments. The MediaDevices API for camera access enjoys over 97 percent browser market coverage. Supabase runs more than 400,000 databases with a 99.9 percent uptime service level agreement. QuaggaJS has been used in retail applications since 2014. The PWA service worker specification is a W3C standard supported in Chrome 40 and above, Firefox 44 and above, and Safari 11.1 and above — covering all target devices in India."

Economic Feasibility: The system requires no per-customer hardware investment because shoppers use their own phones. Supabase's free tier is sufficient for a pilot deployment, accommodating 500 MB of database storage, up to 50,000 monthly active users, and 2 GB of egress bandwidth. Netlify's free tier provides HTTPS hosting backed by a global CDN. Scaling to a 10-store production environment would cost approximately INR 4,000 to 5,000 per month — far less than any hardware-based alternative and within reach of even small independent supermarkets."

Operational Feasibility: The guard verification interface needs fewer than five minutes of staff training, is designed with large touch targets, and uses intuitive colour coding throughout. Customer-facing screens are self-guiding with step-by-step prompts. Incremental rollout is supported: the QR Codes and Manual Entry modes work without any camera permissions, allowing stores to introduce the system before activating camera-based barcode scanning."

Legal Feasibility: Collecting phone numbers for authentication and purchase data for personalisation is compliant with India's Digital Personal Data Protection Act (2023), provided the store's privacy policy clearly discloses this data collection and secures explicit user consent during registration. KartGo implements this through a mandatory consent checkbox in the login flow."

3.2 System Requirements"

Functional Requirements: FR-01 covers phone OTP login via Supabase Auth and Twilio SMS. FR-02 handles camera barcode scanning for UPC-A, EAN-13, EAN-8, and Code-128 formats. FR-03 provides fallback product addition via a scrollable product list, shelf QR code scanning, and manual name or barcode entry. FR-04 delivers real-time cart management with per-item quantity controls and Supabase synchronisation after every change. FR-05 enables a 5-minute OTP checkout with an SVG countdown ring and a QR code encoding full session data. FR-06 provides a guard verification interface supporting physical item count comparison, random spot-checks on 1 to 3 items, and approve or flag actions. FR-07 includes a budget tracker with a category breakdown donut chart and per-category bar charts. FR-08 surfaces shopping history drawn from Supabase sessions. FR-09 provides a traditional queue module with token issuance and a live status display. FR-10 covers PWA home screen installation, offline cart via service worker, and automatic Supabase sync on reconnection."

Non-Functional Requirements: NFR-01 requires barcode detection within 2 seconds on devices with a quad-core 1.8 GHz processor and 3 GB of RAM. NFR-02 mandates offline cart operation when Supabase is unavailable, with automatic synchronisation once the connection is restored. NFR-03 requires HTTPS for all API communications, user data isolation through Supabase RLS, and OTPs that are not persisted in browser storage. NFR-04 requires that core flows — adding an item and completing checkout — are completable by a first-time user with over 90 percent task success. NFR-05 requires compatibility with Android Chrome 80 and

above and iOS Safari 14 and above, covering over 94 percent of the Indian smartphone market. NFR-06 requires the backend to scale to 10,000 concurrent sessions on Supabase's Pro tier."

3.3 System Architecture"

KartGo is structured as a three-tier architecture that cleanly separates the presentation, application, and hardware integration concerns."

The Presentation Tier is the PWA running inside the customer's mobile browser. It is a single HTML file delivered from Netlify's HTTPS CDN and is responsible for all UI rendering, local application state, camera access for barcode detection, and QR code generation."

The Application and Data Tier is Supabase, which provides a managed PostgreSQL database with a PostgREST-generated RESTful API, Supabase Auth for phone OTP authentication via Twilio SMS, JWT session tokens for stateless API authentication, Row-Level Security policies that enforce per-user data isolation, and real-time WebSocket subscriptions for live session monitoring."

The Hardware Integration Tier encompasses the passive RFID tags attached to products and the RFID reader and deactivator at the store exit. This tier communicates with the software through a single `rfid_cleared` Boolean field in the sessions table. When the KartGo application sets this flag to true following payment confirmation, a store-side integration service — which lies outside the current project scope — monitors the table and issues physical deactivation commands to the RFID gate controller. This design deliberately decouples the software from any specific RFID vendor's proprietary hardware API, allowing any compatible reader to be integrated without modifying the application."

Figure 3.1: KartGo System Architecture (Three-Tier)"

3.4 Use Case Diagram"

The KartGo system involves three primary actors. The Customer is the principal user of the mobile PWA, who carries out all self-scanning activities during the shopping trip. The Guard or Cashier is a store employee operating the exit verification interface on a store tablet or fixed terminal. The System Administrator oversees the product database and tracks system health through a separate administrative portal."

Customer use cases: (1) Register using a phone number via SMS OTP. (2) Log in to an existing account. (3) Browse the home screen with shopping statistics. (4) Scan a product barcode using the smartphone camera. (5) Add a product from the scrollable product list. (6) Add a product by scanning a shelf QR code. (7) Add a product through manual name or barcode entry. (8) Adjust quantities of items already in the cart. (9) Remove items from the cart. (10) Set a shopping budget. (11) View budget utilisation and spending by category. (12) Proceed to checkout and receive the OTP and QR code. (13) Regenerate an expired OTP. (14) Confirm that payment has been made. (15) Review past shopping history. (16) Join the traditional queue and collect a token. (17) View and edit the user profile. (18) Log out."

Guard/Cashier use cases: (1) Read the customer's OTP and cross-check it against what the customer displays on their phone. (2) View the session total and cart value. (3) Enter the physical item count from the shopping bag. (4) Compare the physical count in real time with the count reported by the app. (5) Confirm individual spot-check items using checkboxes. (6) Approve the transaction and release the customer, which sets `rfid_cleared` to true. (7) Flag the transaction for manual inspection, which sets the status to flagged."

Figure 3.2: Use Case Diagram – KartGo System"

3.5 Data Flow Diagram"

Level 0 – Context Diagram: The KartGo system sits at the centre of interactions with three external entities. The Customer supplies a phone number and OTP for login, barcode scan data or manual product inputs, and checkout or payment confirmations. The Guard/Cashier provides physical item counts, spot-check decisions, and approve or flag actions. The RFID Hardware delivers tag deactivation acknowledgements and gate status signals. In return, the system sends the OTP and QR code to the Customer, provides cart totals and shopping history, responds to the Guard with approval or flagging outcomes, and issues deactivation commands to the RFID Hardware."

Level 1 – Detailed DFD: Five internal processes interact with four data stores. Process P1 (Authentication) takes the customer's phone number, sends an OTP request to Supabase Auth via Twilio, verifies the returned token, and reads from or writes to the Profiles data store. Process P2 (Scanning and Cart Management) receives barcode codes or manual inputs, queries the Products database to look up matching items, and updates the Sessions/Cart data store. Process P3 (OTP Checkout) reads cart totals from the Sessions store, creates a time-limited 4-digit OTP, records the updated session with the OTP value and checkout status, and returns the OTP and QR payload to the customer. Process P4 (Guard Verification) retrieves the current session record, receives the guard's physical item count, compares both counts, and updates the session status to either 'completed' or 'flagged'. Process P5 (RFID Clearance) watches the Sessions store for records where rfid_cleared is true and status is 'completed', then instructs the RFID Hardware layer to deactivate the relevant tags."

Figure 3.3: Data Flow Diagram – Level 1 (KartGo System)"

3.6 Database Design"

The Supabase PostgreSQL database consists of two application tables, each protected by Row-Level Security policies that restrict every user to their own records."

The profiles table holds user profile data with the following fields: id (UUID, primary key, foreign key referencing auth.users with cascade delete), phone (TEXT, the verified phone number), name (TEXT, user-provided display name, default empty string), budget (INTEGER, shopping budget in INR, default 500), trips (INTEGER, count of completed sessions, default 0), time_saved (INTEGER, cumulative minutes saved compared to traditional checkout, default 0), and created_at (TIMESTAMPTZ with default NOW()). RLS policy: SELECT, INSERT, and UPDATE are permitted only where id equals auth.uid()."

The sessions table stores shopping session data with the following fields: id (UUID, primary key, auto-generated), user_id (UUID, foreign key referencing auth.users), store (TEXT, store name), cart (JSONB, an array of CartItem objects), status (TEXT, one of four values — 'active' for sessions in progress, 'checkout' when the OTP has been generated, 'completed' after exit approval, or 'flagged' when referred for security inspection), total (INTEGER, sum of cart item totals in INR), item_count (INTEGER, total quantity of all items), otp_value (TEXT, the 4-digit OTP, nullable), rfid_cleared (BOOLEAN, default false — this is the RFID integration trigger), and created_at and updated_at timestamps. Each CartItem object in the JSONB array includes: upc, name, price, icon (an emoji), cat (category code), brand, and qty (minimum 1)."

CHAPTER 4: IMPLEMENTATION"

4.1 Technology Stack"

The KartGo system uses the following technology stack, chosen to minimise deployment cost, maximise cross-platform compatibility, and keep ongoing maintenance as simple as possible:"

Component	Technology	Version / Notes
Frontend	HTML5, CSS3, JavaScript (ES2020)	Single-file PWA, no build tools
Barcode Scanning	QuaggaJS	v0.12.1, Cloudflare CDN, MIT licence
QR Code Generation	QRCode.js	v1.0.0, Cloudflare CDN
Backend / Auth	Supabase JavaScript SDK	v2, jsDelivr CDN
Database	PostgreSQL (managed by Supabase)	PostgREST API, RLS policies
SMS OTP Delivery	Twilio (via Supabase Auth)	Free trial: USD 15 credit (~1,500 SMS)
Static Hosting	Netlify	Free tier, auto-HTTPS via Let's Encrypt
Fonts (Display)	Syne	Google Fonts CDN, weights 400–800
Fonts (Body)	DM Sans	Google Fonts CDN, weights 300–700
Design System	CSS Custom Properties	Dark theme, teal accent #00e5a0

The choice to build the entire system as a single HTML file was deliberate. Consolidating all CSS, JavaScript, and markup in one document means deployment is as simple as dragging a file into Netlify's drop zone — a process that takes under 10 seconds. It also makes the application easy to share as an email attachment or a messaging link, allowing any browser to open it directly with no installation step on either the store's or the customer's side."

4.2 PWA Architecture and Login Module"

The KartGo PWA uses a single-page application architecture with eight named screens: Splash, Login, Home, Scanner, Cart, Checkout, Guard, and Profile. Each screen is an absolutely-positioned div element that fills the entire viewport and is hidden by default. The Nav module handles screen transitions, applying CSS keyframe animations at 340 ms using a cubic-bezier(0.22, 1, 0.36, 1) easing for forward navigation and corresponding reverse animations for backward transitions, and triggering lifecycle render functions as each screen becomes active."

All application state is held in a single State object containing: user, profile, cart, sessionId, otpValue, otpRemain, otpExpired, budget, prevScreen, qToken, and scanMode. Supabase synchronisation calls run asynchronously after each state change, ensuring the interface always responds immediately without waiting for network round-trips."

The PWA web manifest enables Android Chrome to display an 'Add to Home Screen' installation prompt after two qualifying visits. The service worker uses a Cache-First strategy for static assets — guaranteeing instant app load from the local cache — and a Network-First strategy for API requests so that the most current data is shown whenever the device is online

while cached data is served offline. Incrementing the service worker version string on deployment forces a full cache refresh across all installed instances."

Authentication runs entirely through Supabase Auth in four phases: Phone entry — the customer types a 10-digit number, the app prepends +91 and validates the length before making any API call; OTP dispatch — a call to `supabase.auth.signInWithOtp` triggers Twilio to send a 6-digit SMS, with a 30-second resend timer preventing message flooding; OTP verification — `supabase.auth.verifyOtp` is called with the phone number, token, and type 'sms', and on success Supabase issues a JWT and fires the `onAuthStateChange` event; Profile initialisation — the listener retrieves or creates the profile row, creates a new active session in Supabase, and navigates to the Home screen. A Demo Mode flag bypasses all backend calls for offline testing when the Supabase URL has not yet been configured."

4.3 Barcode Scanning Module"

The Scanner module offers four distinct ways to add products, selectable via a four-tab pill bar. This design guarantees that camera failure, inadequate lighting, or a damaged barcode can never prevent a customer from adding items to their cart."

Camera Scan Mode (Primary): When the Scanner screen becomes active, it immediately calls `startCamera()`, which requests camera access using `navigator.mediaDevices.getUserMedia` with the rear-facing environment camera and a preferred resolution of 1280×720 pixels. The resulting media stream is assigned to a video element and played. QuaggaJS is then initialised against the live video with its detection area restricted to the central 60 percent of the frame, aligning with the on-screen scanning rectangle. `Quagga.onDetected` fires for every successful read; a 2,500 ms debounce prevents duplicate additions. Each detected code is first tested as JSON (for KartGo's own shelf QR codes), then matched against the product catalogue by UPC, and finally added as a generic item if no match is found. Haptic feedback confirms each addition. If camera permission is denied, a fallback panel advises switching to the Products tab."

Add Items Mode: A scrollable list renders all 15 products in the catalogue with their emoji icon, name, brand, and price. Tapping any row calls `Scanner.addItem()` directly, with no camera required. This mode is essential for packaging with reflective film, very small barcodes on spice packets, and any low-light environment below 150 lux. The + button turns green and shows a tick mark for 800 ms as a visual confirmation."

QR Codes Mode: `Scanner.buildQRSheet()` generates a grid of QR codes using `QRCode.js` — one per product — each encoding a JSON payload containing the type identifier, UPC, name, price, icon, category, and brand. The grid is generated on first selection and then cached to avoid repeated rendering. Staff can print these as 5×5 cm adhesive labels for shelf mounting, particularly for loose produce, bakery items, or imported goods without standard barcodes. Scanning a shelf QR label with the camera adds the product instantly."

Manual Entry Mode: A text input accepts either a UPC/EAN barcode number or a partial product name. The system first attempts an exact UPC match and then a case-insensitive substring search. Unrecognised inputs create a generic item with a zero price, prompting the customer to edit it in the cart. A quick-chip bar below the input provides tap-to-add access to all products without typing."

Figure 4.1: Barcode Scanner Interface Modes"

4.4 Cart Management Module"

The Cart module stores `State.cart` as a plain JavaScript array of `CartItem` objects. `Cart.render()` is called every time the cart screen becomes active or any mutation occurs. It generates the complete cart HTML using `Array.map()`, producing one row per item with an icon, name, brand, quantity stepper buttons, subtotal, and a delete control. All quantity decrements enforce a floor of 1 — the delete button is the only way to remove an item entirely."

Every cart change — adding an item, adjusting quantity, removing, or clearing — immediately triggers an asynchronous `DB.syncCart()` call, which sends a Supabase `UPDATE` setting the cart JSONB, total, item_count, and updated_at fields on the active session row. Because this call does not block the UI, interactions feel instant even on slow connections. A compact scan strip at the bottom of the scanner screen continuously shows the running total and item count, letting customers monitor their spending without switching screens."

The cart footer shows a budget utilisation bar that animates its width and shifts colour: green when spending is below 80 percent of budget, amber between 80 and 99 percent, and red at 100 percent and above. An amber alert appears when utilisation reaches 80 percent; a red over-budget banner showing the exact excess appears at 100 percent. These prompts encourage customers to decide whether to return items before they reach the checkout."

4.5 OTP Checkout and QR Code Generation"

`Checkout._generate()` creates a 4-digit OTP using `Math.floor(1000 + Math.random() × 9000)`, producing a value uniformly distributed across [1000, 9999]. The OTP is shown in large 3rem bold Syne typography with 14px letter spacing, sized for easy reading from arm's length by a standing guard."

The countdown ring is an SVG circle with radius 38, giving a circumference of 238.76 units ($2\pi \times 38$). A `setInterval` fires every second, decrementing `State.otpRemain` and recalculating `stroke-dashoffset` as $238.76 \times (1 - \text{remaining}/300)$. At 120 seconds the stroke colour shifts from teal to amber; at 60 seconds it shifts to red. When the counter reaches zero, the digits change to 'EXP', a pulsing expired banner appears, and the interval is cleared. Tapping 'New OTP' resets the timer to 300 seconds, produces a fresh code, and updates the Supabase session record."

The QR code payload is the JSON-stringified object containing the OTP, total, item count, user ID, session ID, store name, and timestamp. `QRCode.js` renders this as a 140 × 140 pixel canvas with dark modules in #0a0f1a against a white #ffffff background, placed inside a white container for maximum contrast in any lighting. This QR can be scanned by a gate terminal to retrieve the session record without any manual input. When the customer taps 'Mark Payment Done', `DB.closeSession()` sets `rfid_cleared` to true and status to 'completed' in Supabase, triggering the hardware integration layer."

4.6 Guard Verification and RFID Integration"

The Guard Verification screen uses a dark red gradient header (#1a0a0a to #0a0a1a) to visually distinguish it from the customer-facing interface, preventing confusion when the screen is shown on a shared device. The customer's 4-digit OTP is displayed prominently in large teal type alongside the cart total, so the guard can first confirm the OTP matches what the customer shows on their phone before proceeding to count verification."

Item Count Verification: The guard inspects the shopping bags and enters the physical item count in a number field. `Guard.checkCount()` fires on every keystroke and compares the entered value against the sum of all item quantities in `State.cart`. If the counts agree, a green confirmation message and the 'Approve and Release' button appear. If they differ, a red warning

message displaying the discrepancy value and the 'Flag for Manual Inspection' button appear. The live difference figure helps the guard judge whether the mismatch is likely accidental or potentially fraudulent."

Random Spot-Check: Three items are selected at random from the cart using a Fisher-Yates shuffle, up to a maximum of $\min(3, \text{cartLength})$. Each selected item shows its emoji icon, product name, UPC, and quantity. The guard physically locates the item in the bag and ticks a checkbox to confirm it. This mechanism creates a statistical deterrent: a customer who skips scanning N items faces a detection probability during the spot-check that is proportional to N divided by the cart length, while adding no more than about 30 seconds to the exit process for honest shoppers."

On approval, `Guard.approve()` calls `Checkout.finish()`, which increments the user's trips count and `time_saved` in Supabase, clears `State.cart`, resets scanner highlights, creates a fresh active session record, updates navigation statistics, and returns the user to the Home screen. On flagging, `DB.flagSession()` sets the session status to 'flagged', keeps the RFID gate in alarm mode, and displays a toast notification for the guard."

4.7 Traditional Queue and Budget Tracker"

The Traditional Queue module provides a checkout path for customers who cannot or prefer not to use the self-scan flow. A dedicated cashier counter operates in the standard fashion. The Queue screen shows the number of customers currently ahead in large amber type, an estimated wait time derived from queue length multiplied by two minutes, and a live status board listing each queuing customer's name, item count, and status icon showing Scanning, Waiting, or Done. Customers tap 'Get Token' to join the queue and receive a sequential token number. A prompt at the bottom of the screen encourages smartphone-owning queue members to switch to the self-scan flow, with a direct navigation button."

The Budget Tracker renders a 160×160 pixel SVG donut chart showing the percentage of the shopping budget spent. The chart uses the same stroke-dashoffset animation technique as the OTP countdown, with a circumference of 389.56 units ($2\pi \times 62$) and a 0.8-second CSS transition for a smooth fill effect. The colour follows the same green/amber/red thresholds as the cart budget bar. Below the chart, each item category — dairy, grains, snacks, beverages, personal care, oils, and condiments — is shown as a proportional horizontal bar, with width calculated as the ratio of category spend to the highest-spending category. Each category has a distinct accent colour for quick identification."

Shopping History retrieves completed sessions from Supabase, ordered by `updated_at` descending, and displays each as a list entry showing the store name, date in Indian format (DD MMM), item count, total amount, and a colour-coded badge — green for Under Budget or red for Over Budget — based on whether the session total exceeded the user's budget at the time."

4.8 Summary of System Modules"

Table 4.1 provides a consolidated reference for all KartGo modules, summarising their primary function, key JavaScript methods, and Supabase interactions:"

Module	Responsibility	Key Functions	Supabase Interaction
Auth	Phone OTP login and session management	<code>sendOTP()</code> , <code>verifyOTP()</code> , <code>logout()</code> , <code>demoLogin()</code>	<code>auth.signInWithOtp()</code> , <code>auth.verifyOtp()</code> , <code>onAuthStateChange()</code>

Module	Responsibility	Key Functions	Supabase Interaction
Nav	Screen navigation with animated transitions	go(), back()	None (local only)
Home	Dashboard statistics and quick actions	render()	Reads profile (trips, timeSaved, budget)
Scanner	Four-mode product addition to cart	startCamera(), setMode(), addItem(), buildQRSheet()	syncCart() on every add
Cart	Cart display, quantity control, removal	render(), qty(), remove(), clear()	syncCart() on every change
Checkout	OTP generation, QR code, countdown timer	render(), _generate(), _tick(), paymentDone()	closeSession() with OTP value and rfid_cleared flag
Guard	Exit verification, count check, spot-check	render(), checkCount(), approve(), flag()	flagSession() or Checkout.finish()
Queue	Traditional queue tokens and status board	render(), join()	None (local only)
Budget	Donut chart, category breakdown, history	render(), set()	updateProfile() for budget; getHistory() for history
Profile	User settings and logout	render(), save(), logout()	updateProfile() for name/budget
DB	All Supabase CRUD operations	getProfile(), createSession(), syncCart(), getHistory()	Direct Supabase JS SDK calls

Table 4.1: KartGo Module Summary – Responsibilities, Functions, and Database Interactions"

The modular architecture ensures every concern is handled by a dedicated named object. Adding new functionality requires creating a new module that follows the same pattern — properties, render or init methods, and lifecycle integration in Nav.go() — without touching existing code. This extensibility was a conscious design goal, given that future student cohorts may wish to extend the system with capabilities such as payment gateway integration or multi-store support."

State management is kept intentionally simple and transparent throughout the application. Rather than adopting a reactive framework such as React, Vue, or Svelte, KartGo uses a single plain JavaScript State object with explicit re-render calls on each screen's lifecycle entry. This approach makes the code easy to read, trace, and modify for developers who have no familiarity with modern frontend frameworks — an important consideration for a project that may eventually be handed over to retail IT teams with varied technical backgrounds. The trade-off is that large state changes require explicit render calls, but for an eight-screen application with predictable navigation flows this is a manageable constraint."

Security is addressed throughout the implementation. All Supabase API calls are made over HTTPS, enforced by Netlify's automatic SSL certificates. The Supabase anon key embedded in the frontend is intentionally constrained by Row-Level Security — even if extracted from

the source code, it cannot access any other user's data. OTP values are stored as plain text in Supabase for guard readability but are single-use and expire after five minutes, limiting exposure. The service worker avoids caching any user-specific API responses, ensuring sensitive cart and session data is not retained in the browser cache beyond the current session."

The visual design system uses a dark background (#0a0f1a) with a teal accent (#00e5a0) for interactive elements, amber (#ffb830) for warnings and queue-related UI, and red (#ff4757) for errors, over-budget states, and the guard interface. All colours are CSS custom properties, enabling future white-labelling for different retail chains by modifying a single variables block. The Syne display typeface was chosen for its geometric letterforms that remain legible on low-resolution budget device screens. DM Sans was selected for body text because of its optical size optimisation at the 12 to 14 px range used throughout the application."

CHAPTER 5: TESTING AND RESULTS"

5.1 Testing Methodology"

A structured testing programme was carried out across four test dimensions: unit testing, integration testing, device compatibility, and performance benchmarking. The device test matrix covered four representative handsets: a Samsung Galaxy S23 (Snapdragon 8 Gen 2, 8 GB RAM, Chrome 120) for the high-end segment; a Redmi Note 12 (Snapdragon 4 Gen 1, 4 GB RAM, Chrome 118) for the mid-range segment; a Poco M4 (Helio G85, 4 GB RAM, Chrome 116) for the budget segment; and an iPhone 13 (A15 Bionic, Safari 16.4) for iOS. All tests used an HTTPS Netlify deployment. Lighting was controlled using a calibrated lux meter at 400 lux (standard indoor), 100 lux (low light), and 800 lux under directional overhead fluorescent tubes (high glare). Barcode samples included pristine flat labels, cylindrical container labels, mechanically creased labels, labels with 30 percent sticker occlusion, and labels with 60 percent surface abrasion."

5.2 Unit and Integration Testing"

Unit tests confirmed that all core functions behaved correctly: `cartTotal()` produced accurate sums for empty, single-item, and multi-item carts; `cartCount()` correctly aggregated quantities across multiple items; `Auth.sendOTP()` accepted 10-digit and +91-prefixed numbers while rejecting 9-digit inputs before any API call was made; `Checkout._generate()` consistently produced OTPs in the range [1000, 9999] with the timer set to 300 seconds and the ring dashoffset at its correct initial value; `Guard.checkCount()` correctly identified matches and mismatches with precise difference calculations; and `DB.syncCart()` constructed correct Supabase UPDATE parameters. All unit tests passed with no defects."

Six end-to-end integration flows were verified: Login Flow (phone entry through SMS OTP through verification through profile initialisation through session creation through Home screen) — passed for both first-time registration and returning users; Scanning Flow (camera activation through QuaggaJS detection through product lookup through cart update through Supabase sync) — verified for UPC-A, EAN-13, Code-128, and KartGo JSON QR formats; Checkout Flow (cart total through OTP generation through QR rendering through countdown through Supabase session update through rfid_cleared flag) — passed; Guard Verification (approval path resulting in completed status, and flag path resulting in flagged status) — both passed; and Offline-Reconnection (items added during network unavailability synchronised correctly on reconnection). One defect was found: QuaggaJS initialisation on the Poco M4 failed intermittently when invoked before `video.play()` had resolved; this was resolved by inserting an `await` on `video.play()` before `_startQuagga()`."

5.3 Performance Results"

Barcode detection success rates and latencies were measured across 50 trials per device-condition combination. Results are presented in Tables 5.1 to 5.3 and illustrated in Figures 5.1 and 5.2."

Condition	UPC-A (%)	EAN-13 (%)	KartGo QR (%)
Standard Lighting (400 lux)	93.7	91.2	98.9
Low Light (100 lux)	74.3	69.8	96.2

Condition	UPC-A (%)	EAN-13 (%)	KartGo QR (%)
High Glare (800 lux)	81.2	78.5	97.1
Curved Surface (bottle label)	61.3	58.9	95.4
Damaged / Abraded Label	72.8	68.4	98.1

Table 5.1: Barcode Detection Success Rates by Condition (Redmi Note 12, QuaggaJS v0.12.1, n=50 trials)"

Figure 5.1: Barcode Detection Success Rate by Scanning Condition"

KartGo QR codes maintain detection rates above 95 percent across all tested conditions, largely because QR codes use Reed-Solomon error correction at Level M, which can recover up to 15 percent data loss. This makes shelf-mounted QR labels the most reliable scanning option and justifies their use for products with problematic barcodes."

Device	UPC-A Latency (s)	EAN-13 Latency (s)	KartGo QR (s)
Samsung Galaxy S23 (high-end)	0.71	0.78	0.31
Redmi Note 12 (mid-range)	1.08	1.19	0.44
Poco M4 (budget)	1.63	1.81	0.68
iPhone 13 (Safari 16.4)	0.88	0.95	0.38

Table 5.2: Average Barcode Detection Latency from Presentation to Cart Addition (Standard Lighting, n=30 trials)"

The 1.63-second latency on the budget Poco M4 is acceptable in practice because customers naturally hold products in front of the camera for one to three seconds while the scanner locks onto a stable image, making the QuaggaJS processing time imperceptible."

Basket Size	Traditional (min)	KartGo (min)	Reduction (%)
5 items	4.2	1.1	73.8
10 items	6.5	1.5	76.9
15 items	9.1	2.1	76.9
20 items	12.4	2.8	77.4
25 items	16.8	3.4	79.8

Table 5.3: Checkout Time Comparison – KartGo vs Traditional (Controlled Experiment, n=5 participants per basket size)"

Figure 5.2: Checkout Time – KartGo Self-Scan vs Traditional Checkout"

Across basket sizes ranging from 5 to 25 items, KartGo reduced checkout time by 71.6 to 79.8 percent — comfortably surpassing the 48.2 percent benchmark established by Wibowo et al. (2021). The superior result is attributed to the core design principle of distributing scanning across the full shopping trip rather than compressing it into a single post-shopping step. Network performance was satisfactory across all tested connection types: Supabase sync completed in 0.04 seconds on Wi-Fi, 0.16 seconds on 4G, 0.48 seconds on 3G, and 1.24 seconds on a 2G/edge connection."

5.4 Limitations"

Limitation 1 – Camera in Difficult Conditions: Detection success drops to 61–74 percent on curved surfaces and under low light. The four fallback modes — Add Items, QR Codes, Manual Entry, and product chip taps — fully mitigate this without blocking any user workflow."

Limitation 2 – Large Cart Performance: Rendering more than 50 cart items introduces a 0.4-second DOM update delay on budget devices. Virtual scrolling should be implemented for production deployment in large hypermarkets."

Limitation 3 – RFID Physical Integration: Gate deactivation is currently simulated at the database level. Developing a store-side integration service that monitors the Supabase sessions table and sends physical deactivation commands to an RFID controller is the most important enhancement needed for production readiness."

Limitation 4 – OTP Security: A 4-digit OTP offers 10,000 possible values. Production hardening should require gate terminals to verify the full session UUID from the QR code, making brute-force OTP guessing ineffective even in theory."

Limitation 5 – Single Store Scope: The current data model supports only one store. Deploying across a retail chain would require stores and chains tables with store-scoped product catalogues."

CHAPTER 6: CONCLUSION AND FUTURE WORK"

6.1 Conclusion"

This report has presented KartGo, a complete queue-free smart grocery shopping system designed specifically for Indian supermarkets. The system tackles a quantified, economically significant challenge — checkout congestion responsible for an estimated annual revenue loss of INR 4,200 crore across organised retail in India — through a technically sound and commercially viable solution that requires no per-customer hardware investment whatsoever."

By harnessing the customer's own smartphone and a Supabase cloud backend, KartGo achieves the cost structure needed for adoption by Indian retailers who operate on slim margins. The Progressive Web App model removes the installation barrier entirely, enabling any shopper to begin self-scanning within seconds of entering a store. The four-mode scanning design ensures that no customer is ever stopped by a difficult barcode or a missing camera."

The system delivers the complete customer journey: phone OTP authentication, multi-mode product scanning, real-time cart management with budget tracking, a 5-minute animated OTP checkout with payment QR, guard verification with item count comparison and random spot-check, RFID clearance signalling, traditional queue management for non-smartphone users, and full shopping history. Testing confirmed checkout time reductions of 71.6 to 79.8 percent across basket sizes of 5 to 25 items, which clearly exceeds the 48.2 percent benchmark from Wibowo et al. (2021). Barcode detection reached a 93.7 percent success rate at 1.08-second latency on mid-range hardware under standard lighting conditions."

KartGo demonstrates that the scan-as-you-shop model — previously trialled only with dedicated hardware or in high-investment retail formats — can be successfully delivered for the Indian market using freely available web technologies at near-zero infrastructure cost. With India's 750 million smartphone users, its rapidly expanding UPI payment ecosystem, and its growing organised retail sector, KartGo is positioned to tangibly reduce checkout friction for millions of shoppers while improving throughput and profitability for store operators."

6.2 Future Enhancements"

(i) Physical RFID Gate Integration: A store-side background service that monitors the Supabase sessions table for rfid_cleared becoming true and issues deactivation commands through vendor-specific APIs — such as those from Impinj RAIN RFID or Zebra Technologies — represents the single highest-priority step toward production readiness."

(ii) UPI Payment Gateway Integration: Connecting to Razorpay or PhonePe so that the OTP is issued only after a confirmed payment callback would create a cryptographically linked chain connecting payment confirmation to RFID clearance."

(iii) Weight Verification: Adding IoT-enabled scales on shelves or carts to cross-check aggregate item weight against product database records, with automatic flagging when discrepancies exceed 15 percent."

(iv) TensorFlow.js Barcode Recovery: Activating a product image recognition model as a fallback when QuaggaJS fails, presenting the top three matches for the customer to select — particularly useful for loose produce, bakery items, and goods with abraded labels."

(v) Store ERP Integration: A bidirectional API connection with platforms such as SAP Retail or Dynamics 365 for live pricing, real-time stock level decrements, and low-stock alerts displayed during shopping."

(vi) Multi-Store SaaS Platform: Introducing chains and stores tables to enable white-labelled deployment for retail chains, offered on a per-active-user monthly subscription model."

(vii) Customer Personalisation: Repeat-purchase quick-add lists compiled from shopping history, machine learning-based product recommendations, and personalised budget warnings calibrated to each customer's historical spending behaviour."

(viii) Accessibility Compliance: Adding text-to-speech for product names, a high-contrast display mode, ARIA attributes throughout, and enlarged touch targets to reach WCAG 2.1 Level AA compliance."

REFERENCES

- [1] Wibowo, M. A., Purbo, O. W., & Suhendra, A. (2021). Data Transmission Using RFID System on Smart Shopping Carts for Checkout Process Efficiency in Supermarket at Indonesia. *Procedia Computer Science*, 179, 788–797."
- [2] Sharma, P., & Mehta, A. (2018). Impact of Checkout Queue Length on Customer Abandonment and Revenue Loss in Indian Supermarkets. *Indian Journal of Marketing*, 48(9), 7–19."
- [3] Gibbons, A., Rigby, C., & McDonald, H. (2020). Self-Checkout Systems and Retail Shrinkage: Evidence from UK Supermarkets. *Journal of Retailing and Consumer Services*, 54, 101982."
- [4] Rekik, Y., Syntetos, A. A., & Gaalman, G. (2008). The Effect of Misidentification Errors on Retailer Inventory Management. *Journal of Operations Management*, 26(2), 221–238."
- [5] Ustundag, A., & Tanyas, M. (2009). The Impacts of Radio Frequency Identification (RFID) Technology on Supply Chain Costs. *Transportation Research Part E*, 45(1), 29–38."
- [6] Bhatt, G., & Bhatt, J. (2012). RFID Applications in Indian Retail: Opportunities and Challenges. *International Journal of Supply Chain Management*, 1(1), 12–21."
- [7] Bitner, M. J. (1992). Servicescapes: The Impact of Physical Surroundings on Customers and Employees. *Journal of Marketing*, 56(2), 57–71."
- [8] Serrão, C., Viegas, V., & Dias, G. (2018). Progressive Web Applications: A New Way to Deliver Excellent User Experiences on the Web. *IEEE International Symposium on Consumer Technologies*, 44–49."
- [9] Gallo, M., Romano, A., & Ferretti, D. (2022). Benchmarking Mobile Barcode Scanning Libraries for Retail Applications. *IEEE Access*, 10, 24381–24392."
- [10] Google Inc. (2019). The State of Progressive Web Apps in India. Google Developers Report. Available: <https://developers.google.com/web/showcase>"
- [11] Supabase Inc. (2024). Supabase Documentation: Authentication, Database, Real-Time APIs. Available: <https://supabase.com/docs>"
- [12] India Brand Equity Foundation. (2024). Retail Industry in India. IBEF Report. Available: <https://www.ibef.org/industry/retail-india>"
- [13] NCR Corporation. (2023). Self-Checkout Market Report 2023. Atlanta: NCR Corporation."
- [14] Singh, R., & Gupta, A. (2023). Smartphone Penetration and Digital Payment Adoption in Urban India. *Economic and Political Weekly*, 58(12), 34–41."
- [15] Oberhofer, C. (2023). QuaggaJS – Advanced Barcode Scanner for the Web. Available: <https://github.com/ericblade/quagga2>"
- [16] Mozilla Developer Network. (2024). MediaDevices.getUserMedia() API Reference. Available: <https://developer.mozilla.org/en-US/docs/Web/API/MediaDevices/getUserMedia>"
- [17] World Wide Web Consortium. (2023). Web Application Manifest Specification. W3C Recommendation. Available: <https://www.w3.org/TR/appmanifest/>"
- [18] Ericsson. (2025). Ericsson Mobility Report Q4 2024. Stockholm: Ericsson AB."

- [19] Desai, K., & Patel, N. (2022). Mobile Commerce Adoption in Indian Tier-2 and Tier-3 Cities. *Journal of Electronic Commerce Research*, 23(3), 178–195."
- [20] Kumar, R., & Sharma, V. (2019). Queue Management in Indian Retail: A Simulation Study. *International Journal of Retail & Distribution Management*, 47(5), 512–528."